Centering separates residual uncertainty from estimation error

Average the centered expansion Previous slide: pointwise squared-error expansion

$$\|z-a\|^2 = \|r\|^2 + 2r^T(m-a) + \|m-a\|^2$$

Take $E[\cdot\,|y]$; $m-a$ is fixed

$$E[r|y] = E[z|y] - m = 0$$

$$E\left[r^T(m-a)|y\right] = E[r|y]^T(m-a) = 0$$

With $V = \mathrm{Cov}[z|y] = E[rr^T|y]$

$$E\left[\|r\|^2|y\right] = \sum_{j=1}^{d} E\left[r_j^2|y\right] = \operatorname{tr} V$$

Conditional squared-error risk

$$R(a|y) = \operatorname{tr} V + \|m-a\|^2$$

uncover("2-")[- Expected loss is residual uncertainty plus squared distance from the mean.],